

Distance vector Routing algorithm

```
#include<stdio.h>

struct node{
unsigned dist[20];
unsigned from[20];
}rt[10];

int main(){

    int costmat[20][20];

    int nodes,i,j,k,count=0;

    printf("Enter the number of nodes: ");

    scanf("%d",&nodes);

    printf("Enter the cost matrix: \n");

    for(i=0;i<nodes;i++){

        for(j=0;j<nodes;j++){

            costmat[i][i]=0;

            scanf("%d",&costmat[i][j]);

            rt[i].dist[j]=costmat[i][j];

            rt[i].from[j]=j;

        }

    }

    do{

        count=0;

        for(i=0;i<nodes;i++){

            for(j=0;j<nodes;j++){

                for(k=0;k<nodes;k++){

                    if(rt[i].dist[j]>costmat[i][k]+rt[k].dist[j]){

                        rt[i].dist[j]=rt[i].dist[k]+rt[k].dist[j];

                        rt[i].from[j]=k;

                        count++;

                    }

                }

            }

        }

    } while(count>0);

}
```

```

    }
}
}
}
}while(count!=0);
for(i=0;i<nodes;i++){
printf("\n\n FOR router %d\n",i+1);
    for(j=0;j<nodes;j++){
        printf("\t\n node %d via %d Distance %d",j+1,rt[i].from[j]+1,rt[i].dist[j]);
    }
    printf("\n\n");
}
}

```